

Software Requirements Specification

for

Ellmuna App



Software Engineering

Prepared by
Team Members:

#	Name	ID	Role
1	Remas Alshehri	2220005349	Leader
2	Joud Alotaibi	2230006186	Member
3	Jori alghamdi	2230002748	Member
4	Jood Alali	2230004483	Member
5	Raya AL-harbi	2230005278	Member
6	Raghad Alasfoor	2230003688	Member

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Advisor: Dr. Noorah Albehajjan

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Table of Contents

Table of Tables	iii
Table of figures.....	iii
1. Introduction.....	1
1.1 Purpose	1
1.2 Document Conventions	1
1.3 Intended Audience and Reading Suggestions.....	1
1.4 Product Scope.....	2
1.5 References	2
2. Overall Description	3
2.1 Product Perspective	3
2.2 Product Functions.....	4
2.3 User Classes and Characteristics	6
2.4 Operating Environment	6
2.5 Design and Implementation Constraints.....	6
2.6 User Documentation	7
2.7 Assumptions and Dependencies	7
3. External Interface Requirements	7
3.1 User Interfaces	7
3.2 Hardware Interfaces.....	12
3.3 Software Interfaces	12
3.4 Communications Interfaces	12
4. System Features	13
4.1 Home Screen	13
4.2 Weighted Score Calculator (High School Student)	14
4.3 GPA Calculator (University Student).....	15
4.4 Saudi Universities Section.....	16
4.5 Settings	17
5. Use Cases.....	18
5.1 Home Screen	18
5.2 Weighted Score Calculator (High School Student)	18
5.3 GPA Calculator (University Student).....	19
5.4 Saudi Universities Section.....	19
5.5 Settings	19
5.6 Ellmuna Application.....	20
6. Other Nonfunctional Requirements.....	26
6.1 Performance Requirements.....	26
6.2 Safety Requirements.....	26
6.3 Security Requirements.....	26
6.4 Software Quality Attributes.....	27
6.5 Business Rules.....	28
7. Other Requirements	29
7.1 Database Requirements	29
7.2 Legal and Compliance Requirements.....	29
7.3 Reliability and Availability	29
7.4 Usability Requirements	29
Appendix A: Glossary.....	30
Appendix B: To Be Determined List.....	31

Table of Tables

TABLE 1 USER CLASSES AND CHARACTERISTICS.....	6
TABLE 2 USE CASE FOR VIEW HOME SCREEN.....	21
TABLE 3 USE CASE FOR CALCULATE WEIGHTED SCORE.....	22
TABLE 4 USE CASE FOR CALCULATE GPA.....	23
TABLE 5 USE CASE FOR BROWSE SAUDI UNIVERSITIES	24
TABLE 6 USE CASE FOR MANAGE SETTINGS	25
TABLE 7 GLOSSARY OF TERMS.....	30
TABLE 8 SUMMARY OF TO BE DETERMINED (TBD) ITEMS	31

Table of figures

FIGURE 1 ILLUSTRATES THE OVERALL SYSTEM CONTEXT OF THE “ELLMUNA” APPLICATION.	3
FIGURE 2 FUNCTIONAL FLOW DIAGRAMS OF HOME INTERFACE STRUCTURE.....	4
FIGURE 3 FUNCTIONAL FLOW DIAGRAMS OF WEIGHTED SCORE INTERFACE STRUCTURE.	4
FIGURE 4 FUNCTIONAL FLOW DIAGRAMS OF UNIVERSITIES IN SAUDI ARABIA INTERFACE STRUCTURE.	5
FIGURE 5 FUNCTIONAL FLOW DIAGRAMS OF UNIVERSITY GPA CALCULATOR INTERFACE STRUCTURE.	5
FIGURE 6 FUNCTIONAL FLOW DIAGRAMS OF SETTINGS INTERFACE STRUCTURE.....	5
FIGURE 7 PROTOTYPE OF ELLMUNA APPLICATION HOME INTERFACE	8
FIGURE 8 PROTOTYPE OF ELLMUNA APPLICATION DEGREE SELECTION INTERFACE.	9
FIGURE 9 PROTOTYPE OF ELLMUNA APPLICATION GPA SYSTEM SELECTION INTERFACE.	10
FIGURE 10 PROTOTYPE OF ELLMUNA APPLICATION UNIVERSITY GPA CALCULATOR INTERFACE.	11
FIGURE 11 USE CASE DIAGRAM: SHOWING USER ACCESS TO THE FOUR MAIN FEATURES FROM THE HOME SCREEN.	18
FIGURE 12 USE CASE DIAGRAM FOR HIGH SCHOOL STUDENTS.....	18
FIGURE 13 USE CASE DIAGRAM FOR GPA CALCULATION	19
FIGURE 14 USE CASE DIAGRAM FOR SAUDI UNIVERSITIES BROWSING	19
FIGURE 15 USE CASE DIAGRAM FOR SETTINGS SECTION.....	19
FIGURE 16 USE CASE DIAGRAM: ELLMUNA APPLICATION	20

Revision History

Name	Date	Reason For Changes	Version
All members	30 oct	Prepared initial version	0.1
All members	1 nov	Update section 1	0.2
All members	5 nov	Update section 3	0.3
All members	7 nov	Update section 2	0.4
All members	10 nov	Completed final version	1.0

1. Introduction

1.1 Purpose

The purpose of this document is to define, analyze, and document the functional and non-functional requirements for the Ellmuna application, which is considered an academic application designed for students in Saudi Arabia at various levels of education.

This application aims to facilitate students' academic processes throughout, enabling them to calculate their academic average accurately and quickly. This is based on the standards adopted in Saudi universities.

This document contains all application details regarding application functions, interfaces, operational environment, and technical limitations, as well as security and quality requirements to ensure reliable, easy-to-use performance.

1.2 Document Conventions

This Software Requirement Specification (SRS) document adheres to the IEEE 830 standard format. Additionally, each section and subsection is numbered sequentially to enhance clarity and facilitate reference. Moreover, functional requirements are labeled using the format (#-REQ), such as (*REQ-1*, *REQ-2*), whereas non-functional requirements are classified by performance, security, and usability. Thus, bold font indicates section headings, key terms, and italic text is used for examples or notes. As for the bullet points, they are used for list items or list-related points, or requirements. Every requirement is written in brief, direct, verifiable language to maintain consistency and ease of tracking throughout development.

1.3 Intended Audience and Reading Suggestions

This document is for all stockholders who are working on or reviewing the development of the Ellmuna application. It serves as a reference to the following groups:

- **Developers and Designers:** to grasp the system's functional and non-functional requirements and to execute the features correctly.
- **Testers and Quality Assurance Team:** to design and implement test cases that ensure the application complies with the specified requirements.
- **Academic Supervisor:** to evaluate the clarity, completeness, and compliance of the system requirements with educational standards.
- **Project Manager:** Monitor the work progress and ensure alignment with academic goals, risk management, and deliverables.
- **End Users (*students*):** to understand the general purpose of the application and the scope of its academic services.

The document is organized into six major sections:

Section 1: provides an introduction and overview of the system.

Section 2: defines the overall description and context.

Section 3: It outlines the requirements for the external interface, including hardware, software, and communication interfaces.

Section 4: provide a detailed description of all system capabilities and functional necessities.

Section 5: It addresses the non-functional requirements such as performance, security, and software quality attributes.

Section 6: it covers all the other additional requirements, including database, legal, reliability, and usability aspects, followed by explanatory appendices.

1.4 Product Scope

The scope of our project is to develop an application that has four sections:

- 1- High school: In this part, the weighted grade will be calculated by entering high school GPA, Qudrat, and Tahsily. Then, it will be shown the academic grade, e.g., Excellent, Good, Pass, etc., as well as a message that will be written on it, the percentage of possibility of being accepted in Saudi universities.
- 2- Bachelor Students: the GPA will be calculated first by choosing whether it is out of 5 or 4. Then, enter the past credit hours and points, and current credit hours and points. After that, the GPA will be shown with a message if they are 1st honor, 2nd honor, or a message warns them.
- 3- Master student: it will be shown Saudi universities that offer master's programs, so they can select any university and only view the available majors.
- 4- PhD student: same way as the master's section, it will view the university and its majors only for PhD programs

1.5 References

- by the Institute of Electrical and Electronics Engineers, Inc. (1998). *IEEE Recommended Practice for Software Requirements Specifications* IEEE-SA Standards Board.
<http://www.math.uaa.alaska.edu/~afkjm/cs401/IEEE830.pdf>
- Ministry of Education. (2025). المملكة العربية السعودية - وزارة التعليم. Moe.gov.sa.
<https://www.moe.gov.sa/en>

2. Overall Description

2.1 Product Perspective

The "Ellmuna" project is an educational smart system designed for students in Saudi Arabia across various educational levels (high school, undergraduate, and postgraduate).

The project serves as a unified platform that combines academic calculation tools (such as weighted percentage and GPA calculators) with informational browsing features (like exploring Saudi universities and their programs).

The application is a standalone system that uses an internal database to process calculations and display results. In the future, it may be integrated with university databases for live data synchronization.

The user directly interacts with the system by inputting data (grades or percentages), and the system instantly generates results without human intervention. The main purpose is to simplify academic tasks and help students make informed academic decisions with ease and accuracy.

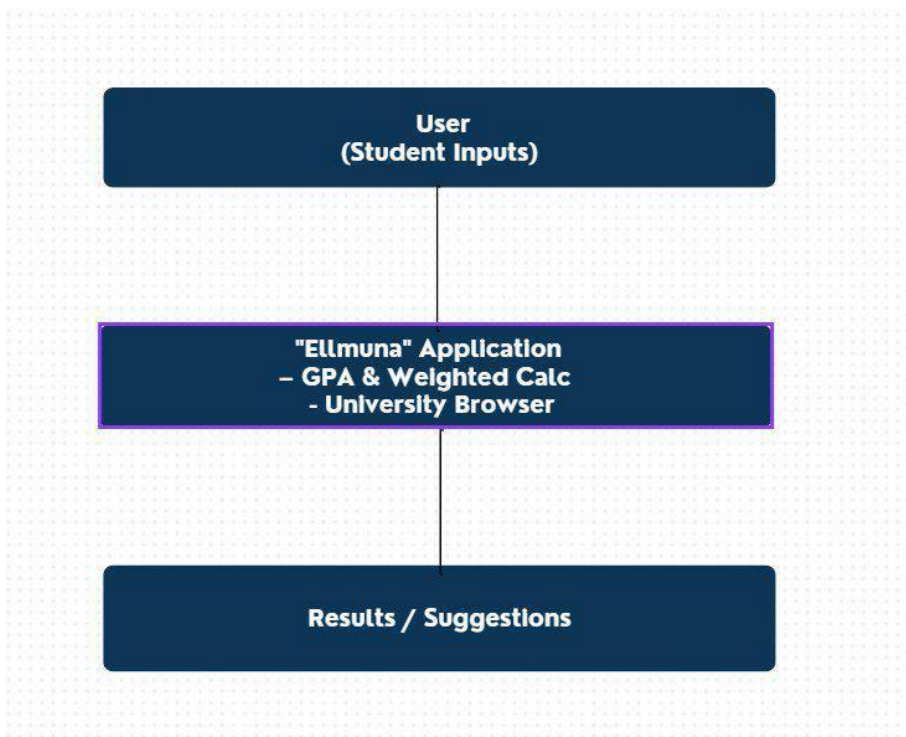


Figure 1 illustrates the overall system context of the “Ellmuna” application.

2.2 Product Functions

The main functions of the “Ellmuna” application include:

1. **Weighted Percentage Calculator (High School Students):**
 - a. Input fields for high school, aptitude, and achievement scores.
 - b. Accurate weighted score calculation.
 - c. Displays a descriptive message reflecting the student’s academic standing.
2. **GPA Calculator (University Students):**
 - a. Supports semester and cumulative GPA calculation.
 - b. Allow users to choose grading scales (out of 4 or 5).
 - c. Displays academic status (e.g., Academic Warning, Honor, Excellent Performance).
3. **University Directory (Bachelor’s, Master’s, and PhD Levels):**
 - a. Provides a list of Saudi universities with their colleges and programs.
 - b. Enables easy browsing, search, and comparison among universities.
 - c. Simplifies academic exploration for all student levels.
4. **Settings Menu:**
 - a. Night mode activation.
 - b. Enable or disable welcome screen.

Interface Home:

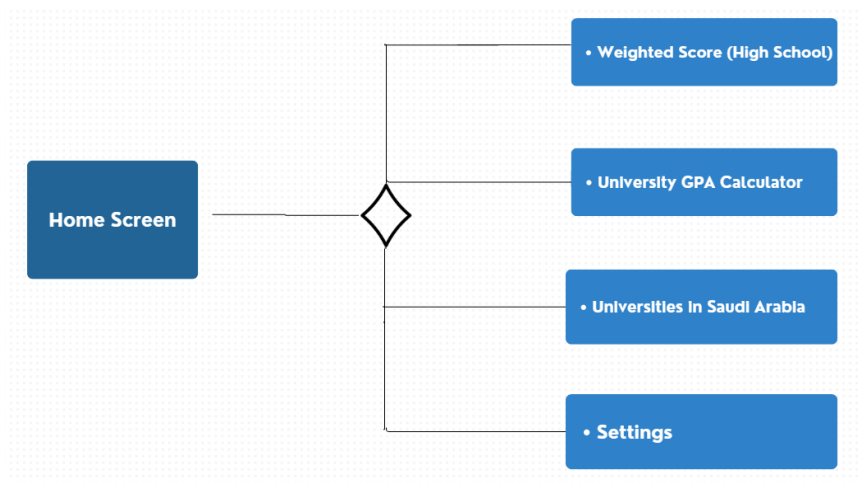


Figure 2 functional flow diagrams of Home Interface structure.

Interface Weighted Score (High School):

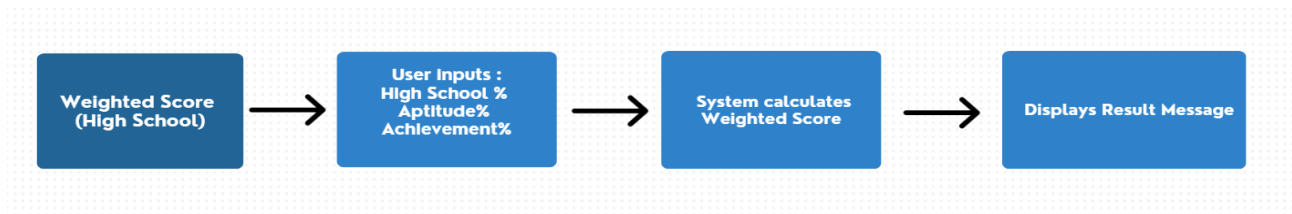


Figure 3 functional flow diagrams of Weighted Score Interface structure.

Interface Universities in Saudi Arabia:

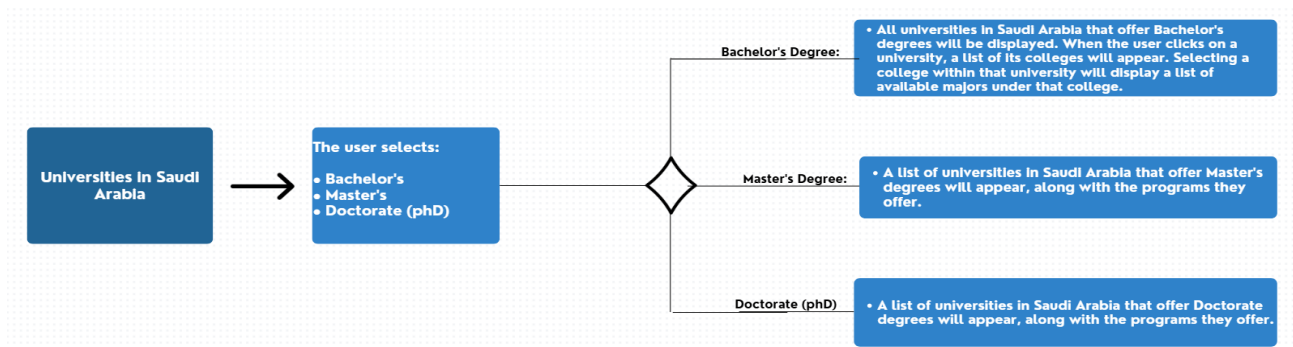


Figure 4 functional flow diagrams of Universities in Saudi Arabia Interface structure.

Interface University GPA Calculator:

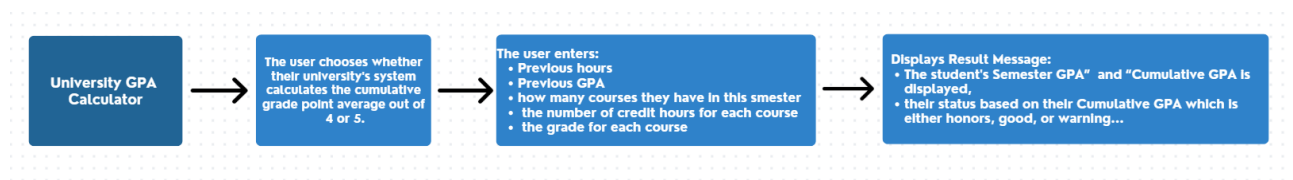


Figure 5 functional flow diagrams of University GPA Calculator Interface structure.

Interface Settings:

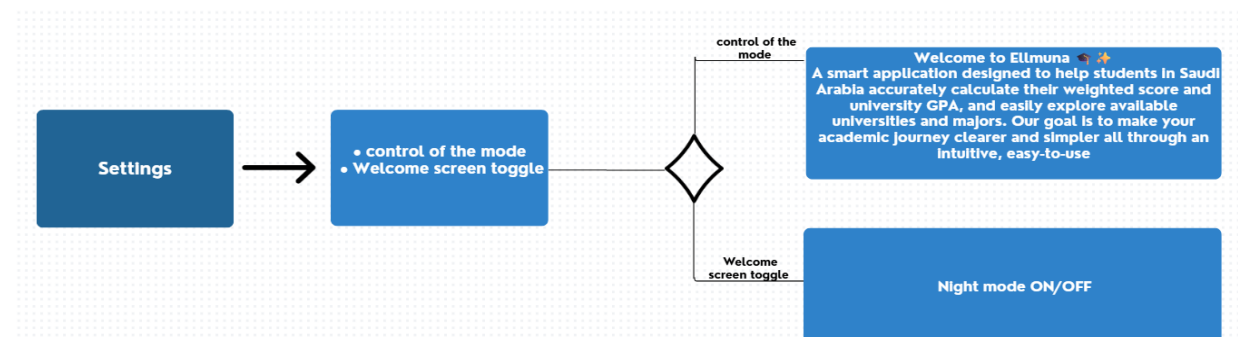


Figure 6 functional flow diagrams of Settings Interface structure.

2.3 User Classes and Characteristics

The application will be used by several user groups:

User Class	Description / Characteristics
High School Students	Primary users use the app to calculate their weighted score and view university eligibility. Have basic mobile skills.
University Students (Bachelor)	Use the GPA calculator to track semester and cumulative GPA. Moderate computer literacy.
Postgraduate Students (Master's/PhD)	Use the app mainly to browse and compare programs offered by Saudi universities.
General Users / Parents	May browse university data for guidance; use the app occasionally.
Admin (Future Role)	May update data or maintain database content in future versions.

Table 1 User Classes and Characteristics

2.4 Operating Environment

The “Ellmuna” application is designed to operate in the following environment:

- **Platform:** Mobile and web platforms.
- **Operating Systems:** Android (version 10 and above) and iOS (version 14 and above).
- **Development Framework:** Flutter or React Native (cross-platform).
- **Backend Server:** Cloud-based hosting environment for storing university data and handling requests.
- **Network:** Requires Internet access for browsing data; GPA and score calculations can be performed offline.

2.5 Design and Implementation Constraints

The following limitations and constraints apply to the system design:

- The system must be implemented using cross-platform mobile development tools (Flutter/Dart).
- The application must support both Arabic (default) and English.
- The user interface must be lightweight and responsive to ensure usability on mobile devices.
- The app will initially operate as a standalone system, with optional integration to APIs for future versions.
- Compliance with Saudi Ministry of Education policies and **PDPL** (Personal Data Protection Law).
- The application must meet accessibility standards for students with varying educational backgrounds.

2.6 User Documentation

The following documentation and help resources will accompany the system:

- **User Manual (PDF):** Explains app installation, usage, and troubleshooting.
 - **In-app Guide:** Step-by-step tips appearing on the first use of each feature.
 - **Online Help Section:** Contains FAQs and short tutorials.
 - **Quick Start Video:** Demonstrates how to calculate GPA or weighted score.
- All documentation will be available in Arabic and English.

2.7 Assumptions and Dependencies

The following assumptions and dependencies are identified:

- Users have access to stable internet and modern mobile devices.
- The default academic weight formulas (30%-30%-40%) can be customized by the user, but the standard values are based on the Ministry of Education's current policy.
- The accuracy of displayed university data depends on updates from external databases or manual input by the team.
- Future integration with official APIs or cloud synchronization services may affect data retrieval performance.
- The system assumes all user-entered data (scores, GPAs) are accurate and valid.

Any modification to grading scales or national admission criteria will require minor system updates.

3. External Interface Requirements

The following section will illustrate the workflow of Ellmuna App interfaces to communication interfaces that show systems, platforms, and operating systems, describing unmatched features.

3.1 User Interfaces

The Ellmuna system includes two primary types of users:

the administrator, responsible for managing and maintaining the system according to organizational guidelines.

And the end user, who interacts directly with the application to access academic tools and information. This section focuses on describing the main interfaces available to the end user.

This is a prototype of some of the interfaces used in the project, intended to illustrate the initial design layout before the final implementation.

3.1.1 Home Page:

This is the main page of the application. It displays the app logo and provides three main options for the user:

- Weighted Average Calculator (High School)
- GPA Calculator (University)
- Saudi Universities

A settings icon is also available at the top left to access app preferences. The user can navigate to any section by selecting one of the buttons.

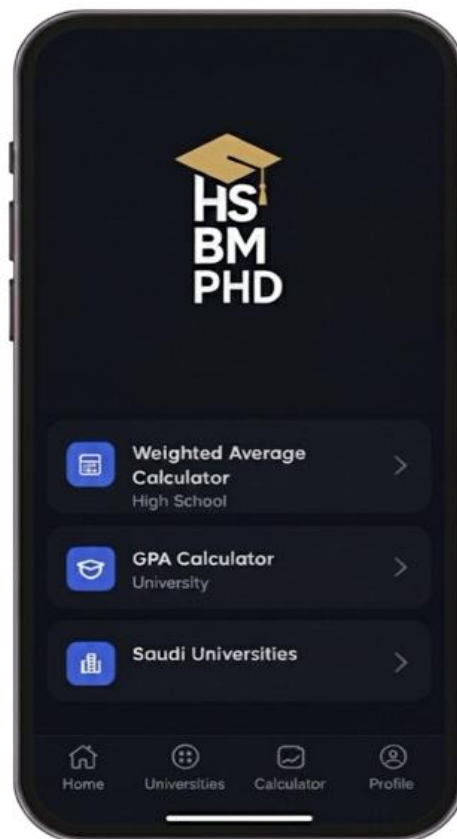


Figure 7 Prototype of Ellmuna Application Home Interface

3.1.2 Degree Selection Page:

This page allows the user to select their degree level in order to filter suitable university programs. The available options are:

- Bachelor's (Undergraduate programs)
- Master's (Postgraduate studies)
- PhD (Doctoral research programs)

The user chooses one degree level, and the app will display matching universities based on the selection.

navigation bar is available at the bottom for quick access to other main sections.

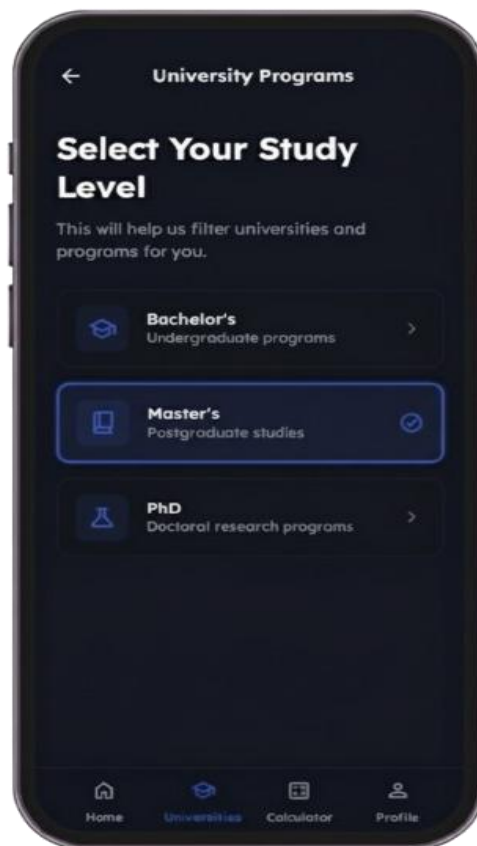


Figure 8 Prototype of Ellmuna Application Degree Selection Interface.

3.1.3 GPA System Selection Page:

This page allows the user to select the GPA scale used by their university before proceeding with the calculation. The available grading systems are:

- GPA out of 4.0 – The standard 4-point scale.
- GPA out of 5.0 – Commonly used by many Saudi universities.

Once the user selects one option, they can click the Continue button to move to the next step.

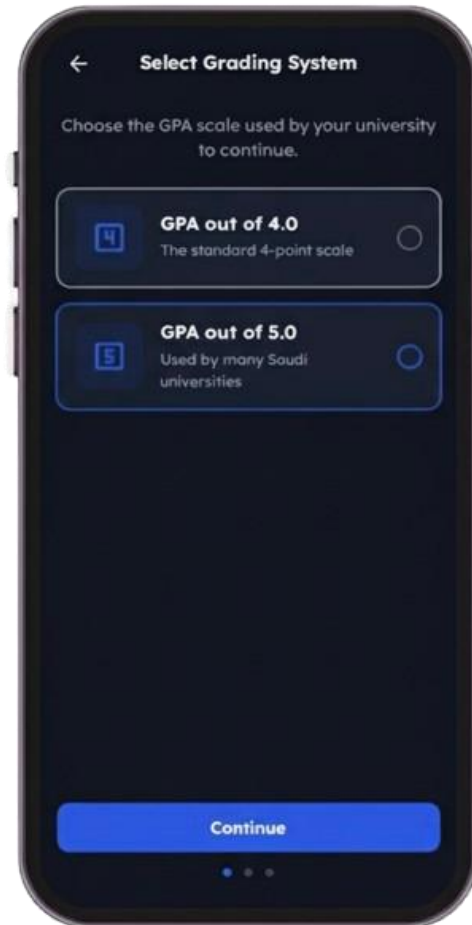


Figure 9 Prototype of Ellmuna Application GPA System Selection Interface.

3.1.4 University GPA Calculator Page:

This page allows the user to calculate their updated GPA by entering both previous academic data and current semester courses. The page is divided into two main sections:

Previous Academic Record:

- Previous GPA input field
- Total previous credit hours input field

Current Semester:

- A list of courses where the user can enter credit hours and select a grade for each course
- An Add Course button to insert additional course rows
- A delete icon to remove any course row

After entering all required information, the user can press the Calculate My GPA button to generate the result.

Previous Academic Record

Previous GPA
e.g., 4.5

Total Previous Credit Hours
e.g., 80

Current Semester + Add Course

Credit Hours	Grade	
3	A+	🗑️
4	A	🗑️

Calculate My GPA

Figure 10 Prototype of Ellmuna Application University GPA Calculator Interface.

3.2 Hardware Interfaces

Ellmuna is a mobile-based system designed to run on Android and iOS devices.

The application requires a cloud server to host the academic data and respond to user requests.

The server should support multi-core processing, SSD storage for fast data retrieval, and high-availability internet access to ensure smooth operation.

End users must have a mobile device with a stable internet connection (Wi-Fi or 4G/5G) to access real-time university information and perform academic calculations.

3.3 Software Interfaces

Ellmuna is compatible with mobile operating systems including Android and iOS and will be developed using a cross-platform framework such as Flutter or React Native.

The system does not require user authentication, and therefore no login or database storage for user accounts is needed.

The application retrieves academic and university-related information from an online API and performs GPA and weighted score calculations within the mobile application.

No MySQL, JDBC, or relational database integration is required for this version of the system.

3.4 Communications Interfaces

The mobile application and the server will communicate securely over HTTPS using TLS encryption to ensure data integrity and confidentiality.

All data exchanged between the application and the cloud server will follow RESTful API standards and use JSON format.

Since no personal data or login credentials are transmitted, communication is strictly limited to retrieving academic datasets and sending calculation requests when needed.

4. System Features

This section describes the main functional features of the Ellmuna application. Each feature outlines a service provided to users, its intended purpose, and the system behavior, including the functional requirements needed for the feature to operate correctly.

4.1 Home Screen

4.1.1 Description and Priority

The Home Screen is the first interface that appears when the user launches the Ellmuna application. It serves as the main navigation page where users can access all major features of the app. This feature is of High Priority because it represents the core entry point and the foundation of user interaction.

4.1.2 Stimulus/Response Sequences

- When the user opens the application, the system displays the Home Screen.
- The interface shows the application logo and name.
- Below the logo, four main buttons are displayed:
 1. Weighted Score Calculator – High School Student
 2. GPA Calculator – University Student
 3. Saudi Universities
 4. Settings
- When the user selects any button, the system redirects them to the corresponding feature page.

4.1.3 Functional Requirements

REQ-1: The system shall display the Home Screen immediately after launch.

REQ-2: The system shall display the logo, name, and four main buttons.

REQ-3: Each button shall correctly redirect the user to the related feature page.

REQ-4: The interface shall be simple and user-friendly for smooth navigation.

4.2 Weighted Score Calculator (High School Student)

4.2.1 Description and Priority

This feature allows high school students to calculate their weighted admission score using their High School, Aptitude, and Achievement Test results. The feature provides a clear final score and a text-based evaluation message based on the result. This feature is of High Priority as it directly serves a primary target audience of the system.

4.2.2 Stimulus/Response Sequences

- The user selects 'Weighted Score Calculator – High School Student' from the Home Screen.
- The system displays input fields for:
 - High School percentage (default weight: 30%)
 - Aptitude Test score (default weight: 30%)
 - Achievement Test score (default weight: 40%)
- The user enters their values and clicks the 'Calculate My Weighted Score' button.
- The system computes the final weighted score and displays the result with an evaluation message.
- The message changes according to the result range (e.g., Excellent, Very Good, Good, etc.).

4.2.3 Functional Requirements

REQ-5: The system shall allow users to input three academic scores.

REQ-6: The system shall use default weight values that can be adjusted.

REQ-7: The system shall calculate the final weighted score using the given formula.

REQ-8: The system shall display a text-based evaluation message according to the calculated score.

REQ-9: The system shall handle invalid or missing inputs by showing an error message.

4.3 GPA Calculator (University Student)

4.3.1 Description and Priority

This feature allows university students to calculate both their semester and cumulative GPA. The feature supports both 4.0 and 5.0 grading systems and provides the student with an academic standing message based on their results. This feature is of High Priority because it is essential for academic progress tracking.

4.3.2 Stimulus/Response Sequences

- The user selects 'GPA Calculator – University Student' from the Home Screen.
- The system prompts the user to select the grading scale (4.0 or 5.0).
- The user enters:
 - Previous GPA
 - Previous total credit hours
- The user clicks 'Continue' and adds new courses by providing:
 - Credit hours for each course
 - Grade obtained in each course
- When the user clicks 'Calculate My GPA,' the system displays:
 - Semester GPA
 - Cumulative GPA
 - Academic status message (e.g., Honor, Warning, Excellent Performance).

4.3.3 Functional Requirements

REQ-10: The system shall allow users to select between 4.0 and 5.0 GPA scales.

REQ-11: The system shall calculate both semester and cumulative GPA accurately.

REQ-12: The system shall display the results clearly, including academic standing.

REQ-13: The system shall handle invalid input or incomplete data appropriately.

REQ-14: The system shall maintain a temporary session for recalculation purposes.

4.4 Saudi Universities Section

4.4.1 Description and Priority

This feature provides users with a categorized and searchable list of Saudi universities and their programs. It helps users explore available Bachelor's, Master's, and Doctorate degrees across different institutions. This feature is of Medium Priority, as it provides useful but non-critical academic information.

4.4.2 Stimulus/Response Sequences

- The user selects 'Saudi Universities' from the Home Screen.
- The system displays three options:
 1. Bachelor's Degree
 2. Master's Degree
 3. Doctorate Degree (PhD)
- The user selects a degree type to view universities offering that program.
- After selecting a university, the system shows available colleges.
- When the user selects a college, the system displays the list of available majors or programs.
- The user can search for specific universities or programs using the search bar.

4.4.3 Functional Requirements

REQ-15: The system shall categorize universities by degree type (Bachelor's, Master's, Doctorate).

REQ-16: The system shall display universities, colleges, and programs hierarchically.

REQ-17: The system shall provide a search function for quick access to information.

REQ-18: The system shall ensure that displayed data is accurate and up to date.

REQ-19: The system shall enable smooth navigation between pages without reloading delays.

4.5 Settings

4.5.1 Description and Priority

This feature allows users to customize their application preferences. Users can manage display options, such as enabling or disabling the welcome screen and switching between light and dark mode. This feature is of Medium Priority, as it enhances usability and comfort but is not required for core functions.

4.5.2 Stimulus/Response Sequences

- The user selects 'Settings' from the Home Screen.
- The system displays available options:
 - Enable or disable the welcome screen.
 - Enable or disable dark mode.
- When the user toggles any setting, the system applies the change immediately.

4.5.3 Functional Requirements

REQ-20: The system shall allow the user to enable or disable the welcome screen.

REQ-21: The system shall provide a toggle for dark and light mode.

REQ-22: The system shall apply changes immediately after the user updates a setting.

REQ-23: The system shall save the user's preferences for future sessions.

5. Use Cases

5.1 Home Screen

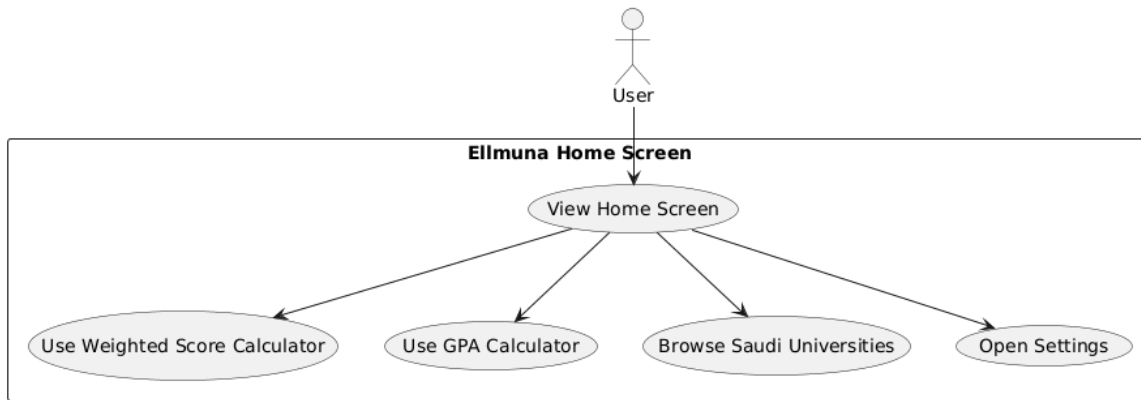


Figure 11 Use case diagram: showing user access to the four main features from the Home Screen.

5.2 Weighted Score Calculator (High School Student)

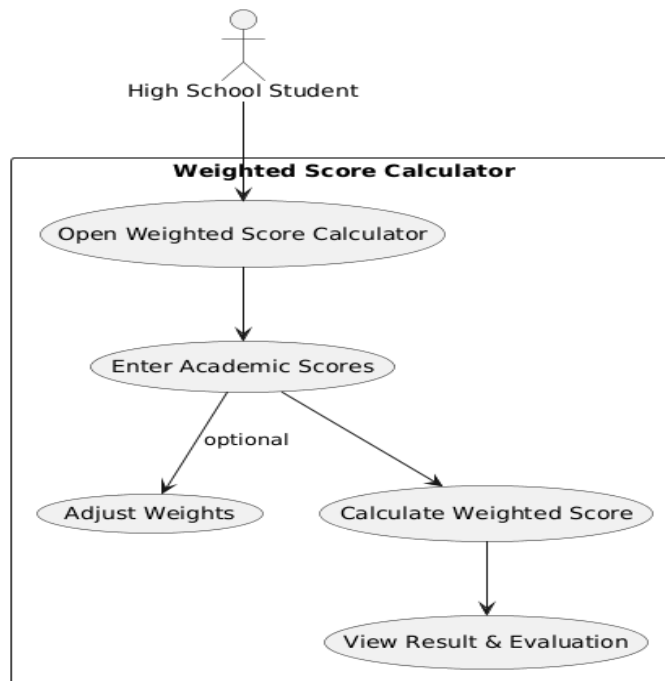


Figure 12 Use case diagram for high school students

5.3 GPA Calculator (University Student)

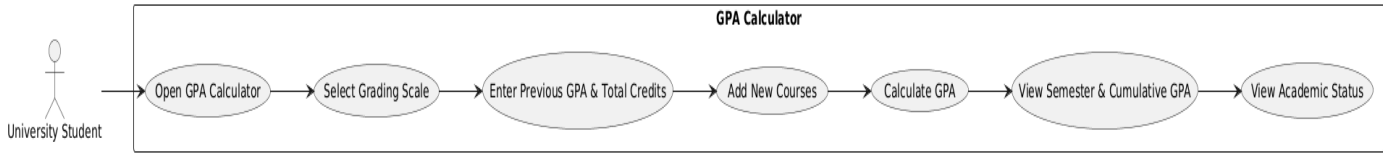


Figure 13 Use case diagram for GPA calculation

5.4 Saudi Universities Section

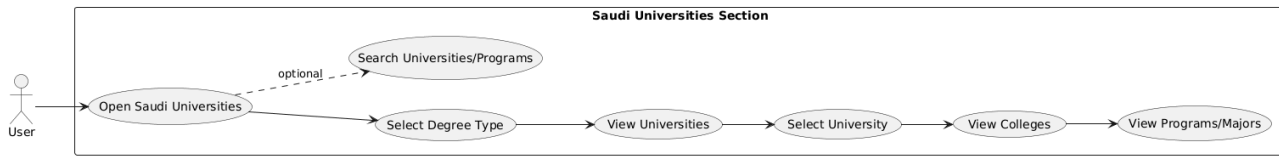


Figure 14 Use case diagram for Saudi Universities browsing

5.5 Settings

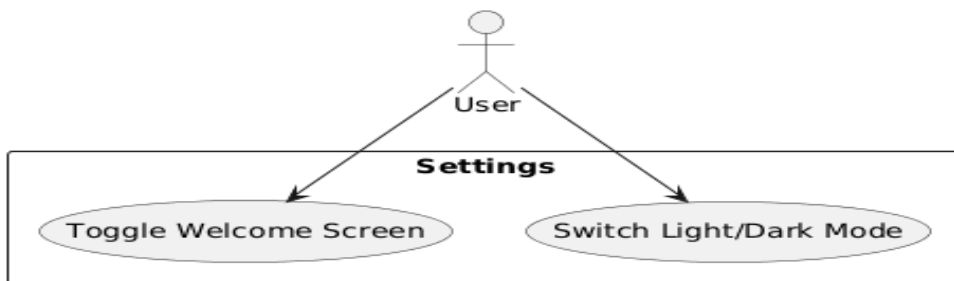


Figure 15 Use case diagram for Settings section

5.6 Ellmuna Application

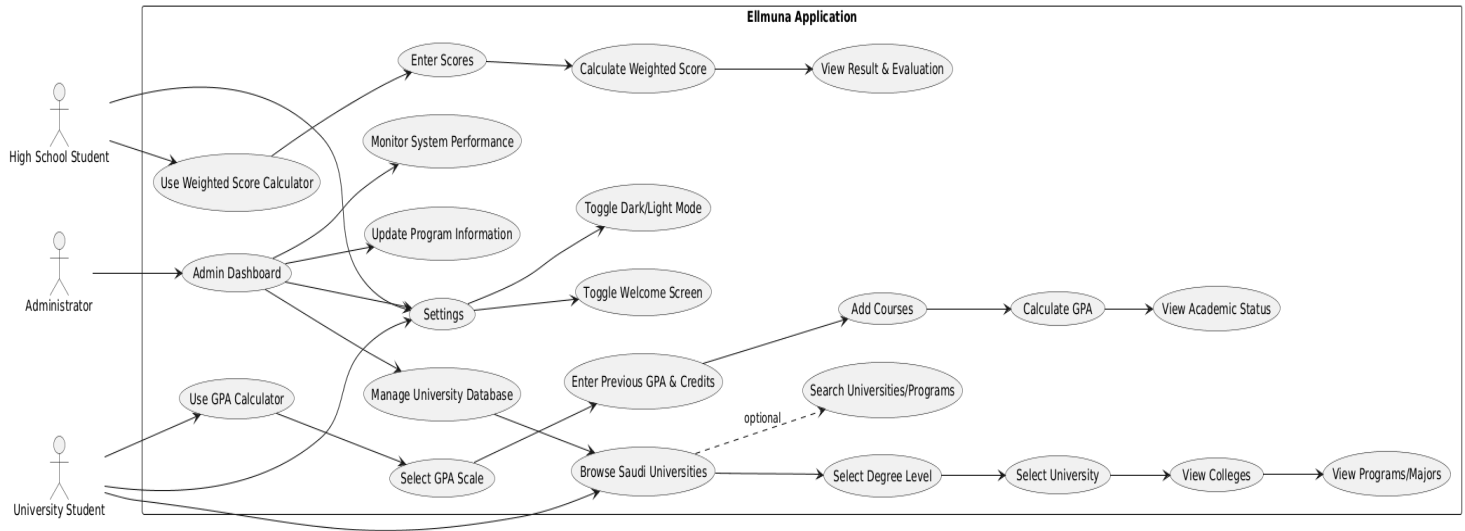


Figure 16 Use case diagram: Ellmuna application

5.7.1 Use Case for View Home Screen

Use Case Name	View Home Screen
Goal	Allow the student to access and view the main interface of the application.
Primary Actors	User
Secondary Actors	System
Preconditions	Application is installed and ready to use.
Postconditions	Home screen with all navigation options is displayed successfully.
Triggers	The use case starts when the user opens the Ellmuna application.

Main Flow	
Step	Action
1	User launches the Ellmuna app.
2	System loads the home interface and displays the app logo.
3	System shows main navigation options.
4	User views available features (Weighted Score, GPA, Universities, Settings).

Extensions	
Step	Action
1	If the system fails to load the home screen, display an error message.

Table 2 Use Case for View Home Screen

5.7.2 Use Case for Calculate Weighted Score

Use Case Name	Calculate Weighted Score
Goal	Enable high school students to calculate their weighted admission score.
Primary Actors	High School Student
Secondary Actors	System
Preconditions	User is on the home screen and selects the Weighted Score option.
Postconditions	Weighted score is calculated and displayed to the student.
Triggers	The use case starts when the student clicks on the Weighted Score Calculator.

Main Flow	
Step	Action
1	Student enters High School, Qiyas, and Tahsili scores.
2	System validates the entered values.
3	System calculates the weighted score based on defined formula.
4	System displays the calculated weighted score.

Extensions	
Step	Action
1	If any field is empty, prompt the user to fill all inputs.
2	If input value is invalid, show an error message.

Table 3 Use Case for Calculate Weighted Score

5.7.3 Use Case for Calculate GPA

Use Case Name	Calculate GPA
Goal	Allow university students to calculate their GPA based on entered grades.
Primary Actors	University Student
Secondary Actors	System
Preconditions	User has opened the GPA Calculator feature.
Postconditions	GPA value is calculated and displayed successfully.
Triggers	The use case starts when the student selects the GPA Calculator from the home screen.

Main Flow	
Step	Action
1	Student inputs course names and grades.
2	System verifies data and credits.
3	System applies GPA calculation formula.
4	System displays the final GPA.

Extensions	
Step	Action
1	If any data is missing, prompt user to enter all required fields.
2	If invalid grade format is entered, system notifies user.

Table 4 Use Case for Calculate GPA

5.7.4 Use Case for Browse Saudi Universities

Use Case Name	Browse Saudi Universities
Goal	Allow users to browse a list of Saudi universities for different degrees.
Primary Actors	User
Secondary Actors	System
Preconditions	User is on the home screen and selects 'Browse Universities'.
Postconditions	List of Saudi universities is displayed to the user.
Triggers	The use case starts when the user clicks on the 'Browse Universities' button.

Main Flow	
Step	Action
1	User selects the degree level (Bachelor, Master, PhD).
2	System fetches the list of universities from the database.
3	System displays the university names and details.

Extensions	
Step	Action
1	If the data fails to load, display 'No connection' message.
2	If no universities are available, show 'No results found'.

Table 5 Use Case for Browse Saudi Universities

5.7.5 Use Case for Manage Settings

Use Case Name	Manage Settings
Goal	Enable users to modify app preferences such as dark mode or notifications.
Primary Actors	User
Secondary Actors	System
Preconditions	User is on the settings page.
Postconditions	Settings are updated successfully and saved in the system.
Triggers	The use case starts when the user opens the settings page.

Main Flow	
Step	Action
1	User selects desired settings (Dark Mode, Language, Notifications).
2	System saves the new settings to the database.
3	System applies the changes instantly.

Extensions	
Step	Action
1	If saving fails, show error message 'Unable to save changes'.
2	If no internet connection, prompt user to reconnect.

Table 6 Use Case for Manage Settings

6. Other Nonfunctional Requirements

Non-functional requirements concern the manner of operation rather than the implementation of functions. They formally define quality attributes and operational constraints of the system, and the intended user experience that are important for ensuring overall performance and reliability.

6.1 Performance Requirements

Performance requirements define the speed, responsiveness, and efficiency that should characterize the functioning of Ellmuna, both under normal and peak conditions.

The system should then calculate and display GPA or weighted scores within 2 seconds once the user inputs the data.

Database queries, such as on university or program information, should be completed in under 3 seconds.

The system should support at least 500 simultaneous users without a decrease in performance and utilize resources such as the CPU, memory, and bandwidth very efficiently.

6.2 Safety Requirements

Safety requirements are defined as the measures and precautions needed to protect the system and the users from potential loss, damage, or misuse.

The system shall ensure that all user data, including GPA, test scores, and personal information, is stored in a secure manner to avoid losses or any other form of damage or corruption.

Back-up procedures should be introduced that would provide for the recovery of data in case the system crashes or suddenly goes out of order.

Safeguards such as confirmation messages before saving or deleting any data need to be put in place so that no academic record gets deleted or altered inadvertently or for unauthorized reasons.

The proposed system shall follow general data protection and safety practices commonly used in educational applications and should be compliant with “PDPL” the Saudi Personal Data Protection Law, regarding proper handling of information.

6.3 Security Requirements

The security-related requirements of the Ellmuna application are about data integrity, assurance of user privacy, and carrying out system operations in a secure way without relying on the login mechanism. Because authentication has not been implemented, internal access control is supported so that changes to university or program data can be affected by an authorized administrator only.

The application should check all user inputs in order to prevent invalid or malicious data entry. Examples of such entries are an invalid GPA or score values. Moreover, if the system will operate online, it is necessary to secure communication between the components with HTTPS to protect the data in transit. The system shall process users' information and educational data responsibly with due consideration to the Saudi Personal Data Protection Law (PDPL) and international privacy standards and to keep this information fully always protected.

6.4 Software Quality Attributes

The Ellmuna application must maintain a high level of software quality to ensure reliability, usability, and long-term maintainability. The following quality attributes define the standards that the system should meet throughout its lifecycle:

6.4.1 Adaptability:

The system should be easily adaptable to future updates, including changes in university admission formulas, grading systems, or academic policies, without requiring a full redesign of the application.

6.4.2 Usability:

The user interface must be intuitive and simple, allowing students to perform GPA and weighted score calculations with minimal training. User feedback should be collected during testing phases to enhance usability and accessibility for all educational levels.

6.4.3 Maintainability:

Developers should be able to modify, update, or correct the system efficiently. The source code must be well-structured, commented on, and organized to facilitate debugging and the addition of new features.

6.4.4 Availability:

The system should be available to users 24/7 with minimal downtime. Scheduled maintenance periods should be announced in advance, and any service interruptions should not exceed 4 hours.

6.4.5 Portability:

Ellmuna must be compatible across multiple platforms, including mobile (Android and iOS) and web browsers, ensuring consistent performance and functionality.

6.4.6 Reliability:

The system should operate without critical failures and maintain data integrity across all user sessions. Automatic data backup mechanisms shall be implemented to ensure recovery in case of errors.

6.4.7 Correctness and Testability:

Each module of the system should be tested against defined requirements to ensure accurate GPA calculations and reliable data retrieval. Unit and integration testing should be conducted to verify that all features perform as intended.

6.5 Business Rules

The Ellmuna system has defined sets of business rules that ensure proper use of its features and maintain data integrity among various user groups. These rules guide the way users interact with this system, as well as the extent of access afforded to each specific role.

6.5.1 User Roles:

- High School Students can input their grades, aptitude, and test scores to calculate their weighted averages and view their university eligibility results.
- University Students can enter semester grades to calculate their GPA and academic standing, view performance reports, and track academic progress.
- Administrators manage system data, including updating university information, program details, and verifying user accounts when necessary.

6.5.2 Data Management Rules:

- The addition, editing, or removal of information about universities, colleges, and programs is the responsibility of the administrators. Users are allowed to make changes in personal or academic data, though confirmation is required at changes being made to avoid accidental editing.
- All calculations, whether of GPA or weighted score, should be performed using only verified and standardized formulas approved by the Ministry of Education.

6.5.3 Operational Rules:

- The system shall track all critical actions, such as data updates or deletions, for accountability and audit purposes.
- Any unauthorized attempt to access restricted data should trigger a system alert and temporary account suspension pending review.

7. Other Requirements

These include all requirements besides the ones mentioned in the previous sections that are necessary for the full and proper functionality of the Ellmuna application. The subsequent subsections define the requirements regarding the database, legal, reliability, and usability to keep the system secure, reliable, and user-friendly.

7.1 Database Requirements

- The system shall have a centralized relational database management system to keep user credentials, academic records, GPA data, and university information.
- It shall ensure referential integrity and avoid redundancy of data. It shall support automatic daily backups.
- Use encryption storage for all sensitive data, such as student accounts and academic results.
- The database must be capable of supporting multi-user access at the same time without any performance degradation.

7.2 Legal and Compliance Requirements

- Only verified academic and university information should be shown on the Ellmuna App from those officially sourced in the Kingdom of Saudi Arabia.
- The application shall comply with local educational laws and the university's intellectual property policies.
- All design, code, and content are owned by the Ellmuna development team and Imam Abdulrahman Bin Faisal University.

7.3 Reliability and Availability

- The application should ensure 99% uptime, except during scheduled maintenance.
- In case of failure, the system should recover automatically without loss of data.
- Recovery Time Objective (RTO) = 2 hours, Recovery Point Objective (RPO) = 24 hours.

7.4 Usability Requirements

- Interface: The interface should be simple, responsive, and intuitive, capable of handling students at different academic levels.
- Design should follow consistent visual standards across Arabic and English interfaces.
- It needs to be accessible on both Android and iOS devices.

Appendix A: Glossary

This appendix provides a glossary of key terms and abbreviations used throughout the Ellmuna System Requirements Specification (SRS).

It is intended to ensure clarity and consistency of terminology for all readers and project stakeholders.

Term	Definition	Stands For
Ellmuna App	Intelligent Academic Assistant application designed for Saudi students at different levels of education.	—
Grade point average, GPA	A numeric measure representing the average performance of a student.	Grade point average
HTTP	HTTP over TLS for confidential and integral data transfer	Hypertext Transfer Protocol Secure
API	Interface for retrieving information related to academics/universities that use online services. Application Programming Interface.	Application programming interface
UI	Screens and other visual aspects the user engage in (Home, GPA, Settings...).	User Interface
UX	Overall ease, clarity, and satisfaction while using the app.	User Experience
RTO	Target maximum time to restore service after failure.	Recovery Time Objective
RPO	Maximum amount of data loss acceptable, measured in time.	Recovery Point Objective
PDPL	Saudi law guiding personal-data handling and privacy compliance.	Personal Data Protection Law

Table 7 Glossary of Terms

Appendix B: To Be Determined List

This appendix outlines the elements and decisions that are still pending finalization in the current version of the Ellmuna Software Requirements Specification (SRS).

Each item listed below includes a brief description of what remains to be determined and the resolution plan to ensure that all open points are addressed before the final release.

Item	Description	Resolution Plan
Integration with Official University APIs	Approval is pending to connect Ellmuna with the Ministry of Education (MOE) or individual university data sources for real-time academic information retrieval.	To be confirmed after prototype evaluation and technical review by the development and academic integration teams.
Hosting Platform	The decision between deploying the system on Firebase Hosting or the University's internal server has not yet been finalized.	To be decided before the deployment phase, following a performance and cost assessment.
Final User Testing Plan	The scope, number of participants, and criteria for user acceptance testing are yet to be established.	To be completed during the Software Testing Plan (STP) phase once the prototype is ready.
Data Synchronization Frequency	The update interval for retrieving university and GPA data (daily, weekly, or on-demand) is not yet defined.	To be finalized after integration testing and consultation with MOE technical representatives.

Table 8 Summary of To Be Determined (TBD) Items